

IIITH PGEE 2024 Memory Based Questions

Aptitude Section

2 comprehensions and related questions 4 each

1 tabular data give with 4 related questions (ques was straight forward only cal was bit lengthy)

Basic geometry like finding area of inscribed shape 2, 2 circle question

- 1) Surface area of cube when diagonal given $10\sqrt{3}$ m.
1. Logical reason based on some pretext. Indiabix practice question were enough.
2. Co ordinate geometry questions related to area of 2D shapes.
3. Counting and probability.
4. Two numbers a and b are in ratio 3:4 their LCM is 192. What is the difference between their squares.

Work & Time:

- 1) A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?
- 2) Train & distance:
A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?
- 3)
Alice : charlie is culprit
Charlie : bob is culprit
Eve: I didn't do it
Bob : Charlie is lying
One of them telling truth. Who is culprit?
- 4) 1 to 9 numbers given make 5 digit number 2 odd followed by 3 even digits
Find the number of way to arrange (without replacement)
- 5) If 1st Jan 2006 was then what was day on 1st Jan 2010?
- 6) two points is given in XY plane, it'll represent to which 3D space?

7)

It is being given that $(2^{32} + 1)$ is completely divisible by a whole number. Which of the following numbers is completely divisible by this number?

- A.) $(2^{16} + 1)$
- B.) $(2^{16} - 1)$
- C.) (7×2^{23})
- D.) $(2^{96} + 1)$

Technical Section

Dbms

normalization based nearly 2 or 3
Query given and asked the relevant

C programming

1. Outputs for the given program nearly 2 or 3
2. What will be output.

```
if printf("PGEE2024")  
{
```
3. `arr = {{6,7,8,9},{10,11,12,13}}`. What is the output for `**arr`.
4. What is the return type for the function call to `calloc`?
5. `int x = - - 10; printf(x);`
- 6.

OS

1. What is the use of dirty bit in page table.
2. Suppose we have a semaphore var `s` initialised to zero. If a thread `t`, calls wait on `s`, what will happen to `t`.
3. 1 ques based on parent and child id process

Network

1. Subnetting question

2. Gate pyq like where we had to find fragmentation offset, more fragment flag, packet size.
3. 2014 (set-3) MTU (same question came but changed the data values.) 2 marks

Algorithm

1. Minimum number of swaps required to sort an array.
2. $T(n) = 2T(n-1) - 1$. $T(0) = 1$.
3. Graph was given, we had to find the maximum flow from a node s to node t .
4. Huffman coding.
5. Find S = the set of all binary search trees possible with four distinct key. And H = the average height of all elements in those trees.

DBMS:

- 1) Which of the following is not a DML command? Drop, Update, Delete, Insert, None of the above.

[1999 : 1 Mark]

2.9 Consider the schema $R = (S \ T \ U \ V)$ and the dependencies $S \rightarrow T$, $T \rightarrow U$, $U \rightarrow V$ and $V \rightarrow S$. Let $R = (R_1 \text{ and } R_2)$ be a decomposition such that $R_1 \cap R_2 = \emptyset$. The decomposition is

- (a) not in 2NF
- (b) in 2NF but not 3NF
- (c) in 3NF but not in 2NF
- (d) in both 2NF and 3NF

[1999 : 2 Marks]

- 2) A relation (A,B,C,D) has following fd $A \rightarrow C$, $B \rightarrow D$? What is the normal form?
- 3) When is an index said to be a clustered index.
- 4) Theoretical ques on group by and having clause.

EM:

1. What is the maxima of the equation $2x^3 - 9x^2 + 12x + 5$ in the interval $[0,3]$.
2. Orthogonal matrices property based question.
3. Mean value theorem.
4. Simple differential equations in two variable, ordinary differential equations, limit.(differential eq 3 ques)
5. $\lim_{\pi \rightarrow 4} \sin(\pi - 4)/\pi - 4$ value?

Probability

1. 1 person is given two numbers between 1 to 25, there is box in which ball numbered 1 to 25, two balls are picked simultaneously, the person wins if he picks same numbered ball, what is probability of winning?
2. A counter is currently displaying digit 8 using seven leds. The probability that one of the led will switch off is p . What is the probability of this 8 becoming a 3.

DM

1. ques based on matching no and chromatic no relation (their definition based)

DL

1. ques on mux like which mux will be used to design this mux
2. Given a 1×4 mux asked their output (
3. D FF given with -ve edge triggered with some initial state asking what will be their next state after given clock pulse